

Lesson 5 — Design Challenge — Spec-Build Review

Matches LEGO Teacher Portal lesson: "Chore Champion".

Objective (I can...): I can take a written spec, plan it, build and code it, and present my solution against the requirements. **CS concept:** Applying sequencing, decomposition, flowcharts, and pseudocode to a specified problem · **Builds on:** Lessons 1-4. *Also serves as flex/catch-up — if earlier lessons ran long, use today to finish and consolidate.*

AI Connection (2 min): Shipping an AI feature means meeting a spec and defending your choices in a review — the same loop you'll run today. *First-principles question:* "If two teams meet the same spec differently, how do we judge which design is better?"

Materials: LEGO CS & AI kit + device per pair, a printed spec card per pair (a short requirements list, e.g., " ≥ 4 ordered actions, uses two different outputs, has a clear start and end"), plan sheets, notebooks.

Model already built (Basics model) — the 30 minutes is plan + code + test, small reconfigs only.

60-minute flow - Warm-up (8): Vocabulary carousel — fast whiteboard prompts: algorithm, sequence, decomposition, flowchart, pseudocode, abstraction, iteration. - **Concept + demo (10):** Hand out the spec. Coach models decomposing the spec into requirements, drafts pseudocode for one, then steps back. Set the expectation: you present *against the spec*. - **Build/Code (30):** Pairs decompose the spec, plan it (flowchart or pseudocode), and build/code a solution that meets every requirement on the built model. Coach coaches with questions, doesn't solve. *Swap at 15 min.* - **Challenge (7):** Add one requirement of your own and update the plan so it still documents the whole program. - **Share + pack-up (5): Spec-build review** — each pair presents their build, pointing to where it satisfies each requirement, and takes one question. **Coach photographs each build for the unit parent note.**

Pair roles: Co-plan; then Driver codes / Navigator checks against the spec; swap at 15 min. (*Trio rotates plan/code/review.*)

Differentiation: *Support* — a simpler spec (3 requirements) with a partial plan. *Extension / fast-finisher* — a spec requiring a repeated action and a decision described in words. *Reteach* — hand them their Lesson 4 pseudocode as a starting scaffold.

Exit check: Pair delivers a program meeting every spec requirement and can point to where each is satisfied. **Progress layer:** Send home the **Unit 1 parent note** ("your student can decompose a problem, plan it, and build it to spec") with the **build photo**. **At-home (optional):** Write a one-paragraph spec for a gadget, then list how you'd test that it meets the spec. ~10 min.